

Part 2 of 5

American Carrier Strike Groups: An Electronic Perspective



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The first part of this article in September issue covered defensive combat operations, air defence warfare, fighter aircraft and AEW aircraft combo (phase I), and air search radars and fighter aircraft combo (phase II). Read on:

Phase III: Operations: Aegis accompaniments

Destroyers and cruisers associated with the carrier are called Aegis ships, which are equipped with a robust command and control (C²) system called Aegis. These ships are more potent due to the presence of superior radars and C² systems, which are based on high-performance computers.

Aegis cruisers are well-suited to conduct air defence warfare due to their superior array of radars. These have 2D long-range air-search radar AN/SPN-49, which is also present in the carrier. This radar is followed by passive electronic phased array radar AN/SPY-1.

Aegis destroyers do not have AN/SPN-49 radar. Hence, these cannot detect threats

from a long range. Aegis cruisers with AN/SPN-49 radars detect threats at a long range and then Aegis C² system lets the threats of interest to be tracked by AN/SPY-1 radar.

AN/SPY-1 multi-functional radars.

These are phased-array radars, which are the latest-generation radars that were introduced in the 1980s and are still state-of-the-art. These have antennae composed of many transmitting and receiving (T/R) elements. Each element radiates and receives RF energy individually, a sort of many tiny antennae arranged into one big antenna. A single source supplies RF energy to all T/R elements through computer-controlled phase shifters.

The radar computer controls the direction of the beam by varying the phase of the RF signal emitted by each T/R element. In time-domain analysis, the signal to each T/R element is delayed for a certain duration by these phase shifters. By suitably varying the phase (suitably delaying), a wavefront of electromagnetic energy is formed in the desired direction. In simple words, a beam is formed. A moving beam is formed by varying the phase accordingly. This technique is called electronically slewing the beam, in contrast to mechanical slewing.

In mechanically-slewed radars, position of the beam is constant with respect to the antenna. So the antenna itself must be slewed for scanning targets. The ability to slew rapidly is called agility. For rapid scanning requirements, the antenna must be slewed rapidly. But there are mechanical limitations beyond which the antenna cannot be slewed.

In AN/SPY-1 phased-array radar, the beam-formation technique provides agility for rapid slewing. Since the phase is varied by a computer, the direction of the beam can be switched in microseconds; there is no need for the antenna to move at all. Because of this, the antenna of this radar is



Fig. 5: SM-2 missile being launched from Aegis cruiser